

### **E3.3 COMPUTER MODELLING, SIMULATION AND CONTROL OF AN ELECTROMECHANICAL SYSTEM**

#### **SUMMARY**

##### **I) Area of Study**

[Mechatronics and Control]

##### **II) Learning Objectives**

By completing this project, students will be able to learn about:

a. Theoretical Models, Principles and Concepts

In this experiment, students will model and evaluate the behaviour of a simple electromechanical system with open and closed-loop controls.

b. Experimental Techniques

Students will be introduced to Matlab<sup>®</sup>, a numerical computing environment, which will help them to model and analyze the behaviour of an electromechanical system with open and closed-loop control.

c. Instrumentation

PC installed with Matlab<sup>®</sup>

d. Data Analysis

- Open and Closed-loop transfer functions;
- Transient plots and damping analyses;
- Bode plots for frequency and phase response.

### E3.3 COMPUTER MODELLING, SIMULATION AND CONTROL OF AN ELECTROMECHANICAL SYSTEM

#### 1. INTRODUCTION

A mathematical model of a control system is usually defined to allow the evaluation of two important aspects of the system behaviour, namely:

- a) *stability*: the output does not diverge in time in response to a finite input;
- b) *performance*: ability to meet the desired specifications.

In this experiment, students will be introduced to Matlab®, a numerical computing environment, which will help them to model and evaluate the behaviour of a simple electromechanical system (a DC motor) with open and closed-loop controls.

#### 2. OBJECTIVES

- A. To determine the open-loop response of a system in the time domain (transient plot) and in the frequency domain (Bode plot);
- B. To design a stable closed-loop control system that will meet the following specifications to a unit step input:
  - i) peak overshoot,  $mp \leq +5\%$ ;
  - ii) settling time,  $ts \leq 10$  s.

Please refer to the terms' definition reported in the Appendix.

#### 3. THEORY

##### 3.1 Modelling

A DC motor is used in a control system where an appreciable amount of shaft power is required. DC motors are either *armature-controlled* (fixed field), or *field-controlled* (fixed armature current). Most DC motors used in Mechatronics are armature-controlled, i.e. they employ a fixed magnetic field, and the control signal is the voltage applied to the armature terminals.

In [Nise, 2011] an example of a servomotor system is given. Its transfer function is:

$$G_m(s) = \frac{\theta_m}{E_a} = \frac{n(s)}{d(s)} = \frac{K}{s(s+\alpha)} \quad (1)$$

where  $E_a$  is the motor armature voltage (input), and  $\theta_m$  is the motor shaft angular position (output). The two parameters,  $K$  and  $\alpha$ , account for the electromechanical properties of the motor, such as the moment of inertia of the rotor, the resistance and inductance of the armatures, etc. The values of  $K$  and  $\alpha$  depend on intrinsic properties of the motor, and can be considered as known, as they are usually provided by Manufacturers.

### 3.2 Open-loop response

A typical open-loop control configuration is shown in Figure 1. The input is the desired motor shaft angle ( $\theta_i$ ). A potentiometer is used to convert the input angle ( $\theta_i$ ) into a voltage ( $k_p \theta_i$ ). Such voltage is then amplified by a non-inverting amplifier by a factor  $k$ . The resulting voltage ( $E_a = k_p k \theta_i$ ) is applied at the armature terminals of the DC motor. This causes the rotation of the shaft. The angular position of the shaft,  $\theta_m$ , is the output.

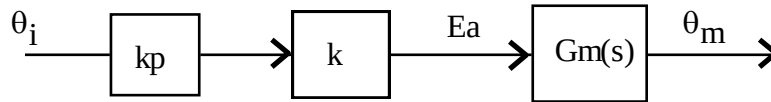


Figure 1: open-loop control.

Considering (1), the open loop transfer function is:

$$G_o(s) = k_p k G_m(s) = \frac{no(s)}{do(s)} = \frac{k_p k K}{s(s+\alpha)} \quad (2)$$

### 3.3 Closed-loop response

Figure 2 shows a closed-loop control system of the DC servomotor. In this case, the output angle ( $\theta_m$ ) is measured by a sensor mounted on the motor shaft (e.g. a potentiometer or an encoder) and it is compared with the desired input angle ( $\theta_i$ ). The position error ( $\theta_i - \theta_m$ ) is converted into a voltage by the potentiometer. The resulting voltage  $k_p (\theta_i - \theta_m)$  is used to drive the motor, after being amplified by a factor  $k$ .

Overall, the voltage applied to the armature of the motor is  $E_a = k_p k (\theta_i - \theta_m)$ . If the output position ( $\theta_m$ ) is equal to the desired position ( $\theta_i$ ), then the position error  $\theta_i - \theta_m$  is zero and no voltage is applied to the motor, whose shaft is therefore prevented from rotating any further.

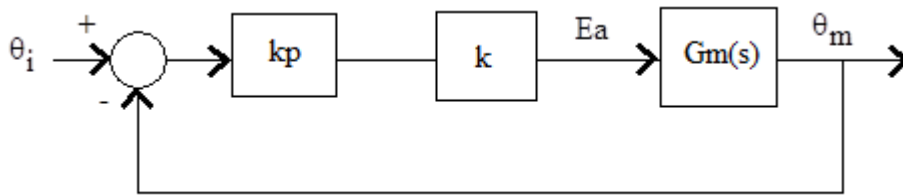


Figure 2: closed-loop control.

The closed loop transfer function is given as [Nise, 2011]:

$$G_c(s) = \frac{\theta_m}{\theta_i} = \frac{nc(s)}{dc(s)} = \frac{k_p k K}{s^2 + \alpha s + k_p k K} \quad (3)$$

Typical questions are:

- is the closed-loop system stable? Please note that the system is considered *unstable* when any bounded commanded position (input) results in an unbounded output, characterized by a continuously increasing position error ( $\theta_i - \theta_m$ ).
- If the system is stable, what value of the amplification factor ( $k$ ) will make the motor turn and stop close to the desired angle ( $\theta_i$ ) within  $t_s$  sec, and without exceeding  $m_p$ ?

## 4. EQUIPMENT USED

PC installed with Matlab

## 5. PROCEDURE

### 5.1 Learning Matlab commands

The following commented example script illustrates the basic Matlab commands needed to perform the analysis of the open-loop and closed-loop control systems of a DC motor. The example transfer function used here has a similar structure as  $G_o$  (2) and  $G_c$  (3):

$$G = \frac{a}{s^2 + bs + a} \quad (a = 20; b = 5)$$

```
clear all    % delete variables in the Workspace
clc         % clear the screen
close all   % close all existing figures, if any

% Let's define the variables:

a = 20;
b = 5;

% Define two variables, n and d, describing the second order polynomials
% respectively at the numerator and the denominator of G, via their
% coefficients:

n = [0 0 a];
d = [1 b a];

% The transfer function (G) is created using the command "tf":
G = tf(n,d);

% Let's compute the unit step response in the time domain.
% The time domain is the interval [tmin, tmax]. The discrete variable "t"
% will span the time domain, assuming the values in the following set:
% { tmin, tmin+incr, tmin + 2 incr ... tmax-incr, tmax }.
% (Note: proper values for tmin and tmax must be defined for each transfer
% function.

tmin = 0;
tmax = 4;
incr = (tmax-tmin)/100;
t = tmin:incr:tmax;

% Unit step response:
step(G,t);grid on

% Let's draw horizontal lines to identify the error band:

error = 0.02; %max error set to 2%, i.e. 0.02
line([tmin,tmax],[1-error,1-error],'Color','red');
line([tmin,tmax],[1+error,1+error],'Color','red');
ylim([0 1.2]); % This sets the limit of the y-axis to 1.2; this limit can
               % be changed as needed for visualization purposes.

% Bode plot:
figure %keep the previous figure (instead of overwriting it)
bode(G); grid on
```

## 5.2 Open loop response

- 1) By adapting the example script, define **no** and **do** for the transfer function of the open-loop system  $G_o$  (eq. 2), using the following numerical values:
  - $k=1$
  - $k_p = 0.862$
  - $K = 0.041$
  - $\alpha = 1.815$
- 2) Obtain the unit step response and the Bode plot of  $G_o$ .
- 3) Print out the screen graphics for each plot (overall: 2 plots).

## 5.3 Closed loop response

- 1) Define **nc**, and **dc** for  $G_c$  (eq. 3). Use the same numerical values for  $k_p$ ,  $K$  and  $\alpha$  as in 5.2. Try different values of  $k$  (in the range 1-100) in order to obtain:
  - a. An under damped response  
(specifications:  $m_p \leq 5\%$ ; settling time:  $t_s \leq 10$  s; error band:  $\pm 2\%$ )
  - b. A critically damped response  
(specifications: settling time:  $t_s \leq 10$  s; error band:  $\pm 2\%$ )
  - c. An over damped response  
(specifications: settling time:  $t_s \leq 10$  s; error band:  $\pm 2\%$ )
- 2) Print the step response plots for each of the three responses above, using suitably defined time scale.
- 3) For each plot, determine the peak value ( $m_p$ ), the peak time ( $t_p$ ), and settling time ( $t_s$ ). Report such values in Table 1. (Please note that Table 1 is already copied in the log sheet).

| Type of response  | k | $m_p$ | $t_p$ | $T_s$ | error (%) |
|-------------------|---|-------|-------|-------|-----------|
| Under damped      |   |       |       |       | 2         |
| Critically damped |   | n/a   | n/a   |       | 2         |
| Over damped       |   | n/a   | n/a   |       | 2         |

Table 1

## 6. RESULTS AND DISCUSSION

### 6.1 Open Loop response

Submit the following plots:

- Step response in the time domain of  $G_o$ ;
  - Bode plots (frequency response) of  $G_o$ .
- a) Explain whether the open-loop system is stable or unstable.
  - b) If the input is a sine signal of amplitude 1 and frequency of 3 rad/s, what is the magnitude and phase of  $\theta_m$ ? (hint: use the frequency response plot).

## 6.2 Closed loop response

Submit the following plots:

- Closed-loop step response for *under-damped condition* (highlighting the numerical values:  $k$ ,  $m_p$ ,  $t_p$ ,  $t_s$ );
  - Closed-loop step response for *critically-damped condition* (highlighting the numerical values:  $k$ ,  $t_s$ )
  - Closed-loop step response for *over-damped condition* (highlighting the numerical values:  $k$ ,  $t_s$ ).
- a) What is your designed value of  $k$  that can meet the specifications (fill in Table 1, reported in the Log Sheet)?
- b) Will  $m_p$  (in the under-damped case) and  $t_s$  (in the over-damped case) increase/decrease if  $k$  is increased?

## 7. CONCLUSION

Did you have a stable open loop system for  $\theta_m$  and were you able to design a closed loop stable positional control system that will meet the desired specifications? Explain.

## 8. REFERENCES

[Nise, 2011] Nise, N. S. (2011). Control system engineering, John Wiley & Sons. Inc, New York.

## APPENDIX

**Settling Time** — the time required by the response curve to enter and stay within the tolerance band centred around the target value.

**Rise Time** — the time required by the response to rise from 10% to 90% of its final value; the faster the response time of the system, the shorter the rise time.

**Peak Time** — the time required for a response to reach the first peak of the overshoot.

**Maximum Overshoot** — the maximum peak value of the response curve measured from the desired position.

**Commanded Position** — the desired target output value;  $\theta_i = 1$  in case of unitary step input.

**Steady-State Position Error** — the error ( $\theta_i - \theta_m$ ) that ultimately occurs when the system is in steady-state conditions.